
EXERCISES

Homework 4

AI503: Calculus

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1 Problem 1

1. Evaluate

$$\int_0^3 (3x + 7) dx$$

as a limit of Riemann sums.

1.1 Solution

We know that the Riemann sums for a single variable is defined as

$$\int_a^b f dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k) \Delta x$$

We know that for each x : $x_k = a + k\Delta x$ and that $\Delta x = \frac{b-a}{n} = \frac{3-0}{n} = \frac{3}{n}$

So therefore

$$x_k = \frac{3k}{n}$$

$$\begin{aligned} \int_a^b f dx &= \lim_{n \rightarrow \infty} \sum_{k=1}^n (3x + 7) \cdot \frac{3}{n} \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(3 \cdot \frac{3k}{n} + 7 \right) \cdot \frac{3}{n} \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{27k}{n^2} + \frac{21}{n} \right) \\ &= \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{27k}{n^2} + \sum_{k=1}^n \frac{21}{n} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{27}{n^2} \sum_{k=1}^n k + \frac{21}{n} \sum_{k=1}^n 1 \right) \end{aligned}$$

We know the formulas: $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ and $\sum_{k=1}^n 1 = n$

So...

$$\begin{aligned}
&= \lim_{n \rightarrow \infty} \left(\frac{27}{n^2} \frac{n(n+1)}{2} + \frac{21}{n} n \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{27(n+1)}{2} n + 21 \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{27(n+1)}{2} n + 21 \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{27n}{2n} + \frac{27}{2n} + 21 \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{27}{2} + \frac{27}{2n} + 21 \right) = \frac{27}{2} + 0 + 21 = \frac{27}{2} + \frac{42}{2} = 34.5
\end{aligned}$$

2 Problem 2

Why does the Fundamental Theorem of Calculus not imply that

$$\int_{-1}^1 \frac{1}{x^2} dx = -\frac{1}{x} \Big|_{-1}^1 = \left(-\frac{1}{1} - \left(-\frac{1}{-1} = -2 \right) \right)$$

2.1 Solution

Lets use the fundemental theorem of falculus on this.

$$\int_a^b f(x) dx = F(b) - F(a)$$

since $\frac{1}{x^2} = x^{-2}$, then $\int \frac{1}{x^2} = \frac{x^{-2+1}}{-2+1} = x^{-1} = \frac{x^{-1}}{-1} = -\frac{1}{x}$

then

$$\int_{-1}^1 \frac{1}{x^2} dx = -\frac{1}{x} \Big|_{-1}^1 = \left(-\frac{1}{1} - \left(-\frac{1}{-1} = -2 \right) \right)$$

This does not make sense because f has a assymtote.

$$f(x) \rightarrow \infty \text{ as } x \rightarrow 0^- \text{ or } x \rightarrow 0^+$$

3 Problem 3

Find a function $g(x)$ such that $g'(x) = \sqrt{1+x^2}$ and $g(2) = 0$.

3.1 Solution

$$\begin{aligned} g(x) &= \int \sqrt{1+x^2} \, dx \\ &= \int \sqrt{1+x(u)^2} \cdot \frac{du}{dx} \, dx \\ &= \int \sqrt{1+\sinh^2(u)} \cdot \cosh(u) \, dx \\ &= \int \sqrt{\cosh^2(u)} \cdot \cosh(u) \, dx \\ &= \int \cosh(u) \cdot \cosh(u) \, du \\ &= \int \cosh^2(u) \, du \end{aligned}$$

since $\cosh^2(u) = \frac{1}{2}(\cosh(2u) + 1)$

$$\begin{aligned} \int \cosh^2(u) \, du &= \frac{1}{2} \int (\cosh(2u) + 1) \, du \\ &= \frac{1}{2} \int \cosh(2u) \, du + \frac{1}{2} \int 1 \, du \end{aligned}$$

4 Problem 4

Find $\frac{d}{dx} \int_{x^2}^3 \frac{\sin(t)}{t} \, dt$.

4.1 Solution

we know that

$$\frac{d}{dx} \int_{x^2}^3 \frac{\sin(t)}{t} \, dt = -\frac{d}{dx} \int_3^{x^2} \frac{\sin(t)}{t} \, dt$$

Use u substitution

$$u = x^2 \quad du = 2x \, dx$$

then

$$-\frac{d}{du} \left(\int_3^u \frac{\sin(t)}{t} dt \right) \cdot \frac{du}{dx}$$

since $\frac{d}{dx} \int_c^x f(t) dt = f(x)$

$$\frac{\sin(t)}{t} dt \cdot \frac{du}{dx} = \frac{\sin(x^2)}{x^2} 2x = \frac{2 \sin(x)}{x}$$

5 Problem 5

True or False? Give reasons for your answer.

- (a) The double integral $\int_R f dA$ is always positive
- (b) If $f(x, y) = k$ for all points in a region R , then $\int_R f dA = k \cdot \text{Area} R$
- (c) If $\int_R f dA = 0$ then $f(x, y) = 0$ at all points of R

5.1 Solution

- (a) Yes, with a function f inside the $\int$ sign anything is possible
- (b) Yes, since f is a constant k , then $k \int_R f dA$
- (c) No, it can be 0 but doesn't have to.

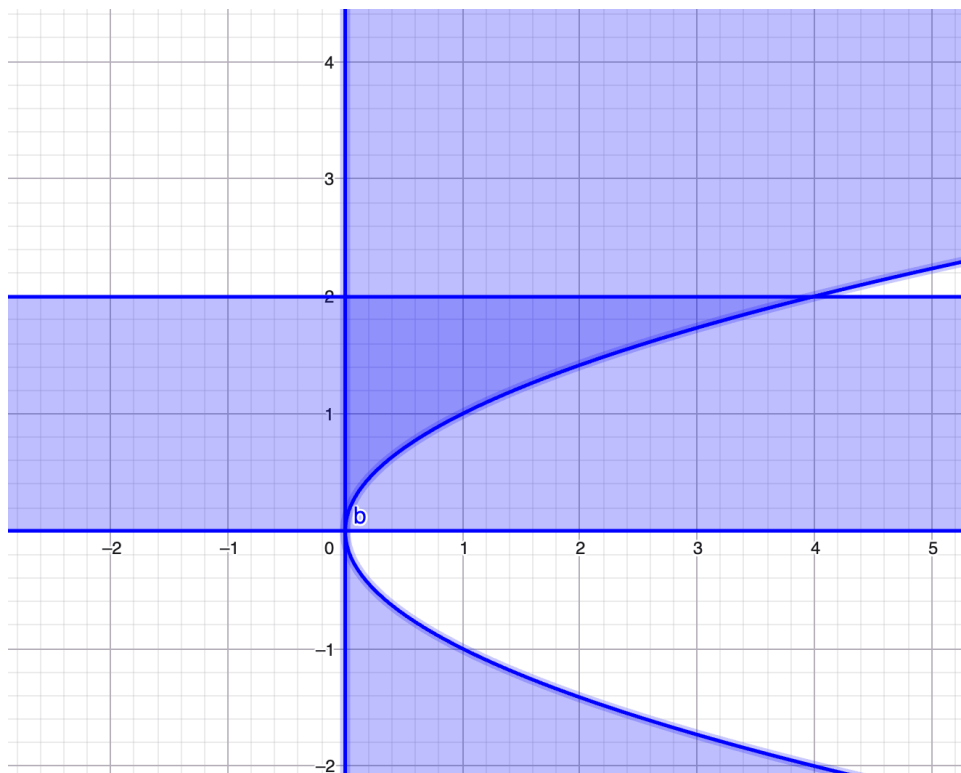
6 Problem 6

Sketch the region of integration

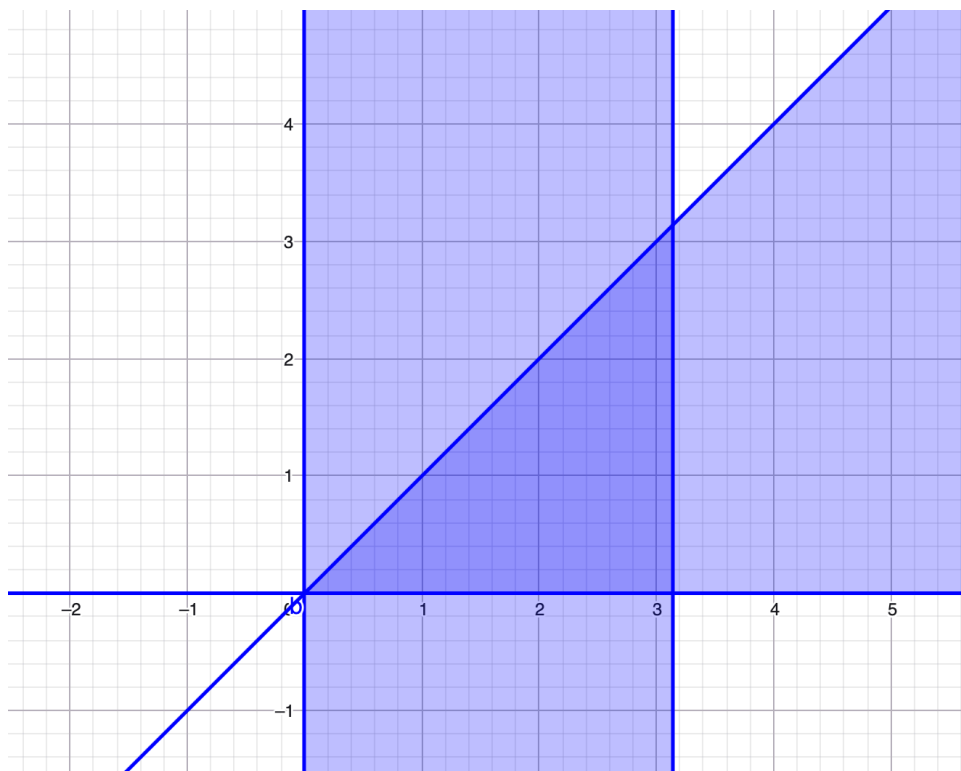
- (a) $\int_0^2 \int_0^{y^2} y^2 x dy dx$
- (b) $\int_0^\pi \int_0^x y \sin(x) dy dx$

6.1 Solution

- (a) We know that $0 \leq x \leq y^2$ and that $0 \leq y \leq 2$



(b) We know that $0 \leq y \leq x$ and that $0 \leq x \leq \pi$



7 Problem 7

A city occupies a semicircular region of radius 3 km bordering on the ocean. Find the average distance from points in the city to the ocean.

7.1 Solution

Pray to allah this will not show up in the exam 🤞

8 Problem 8

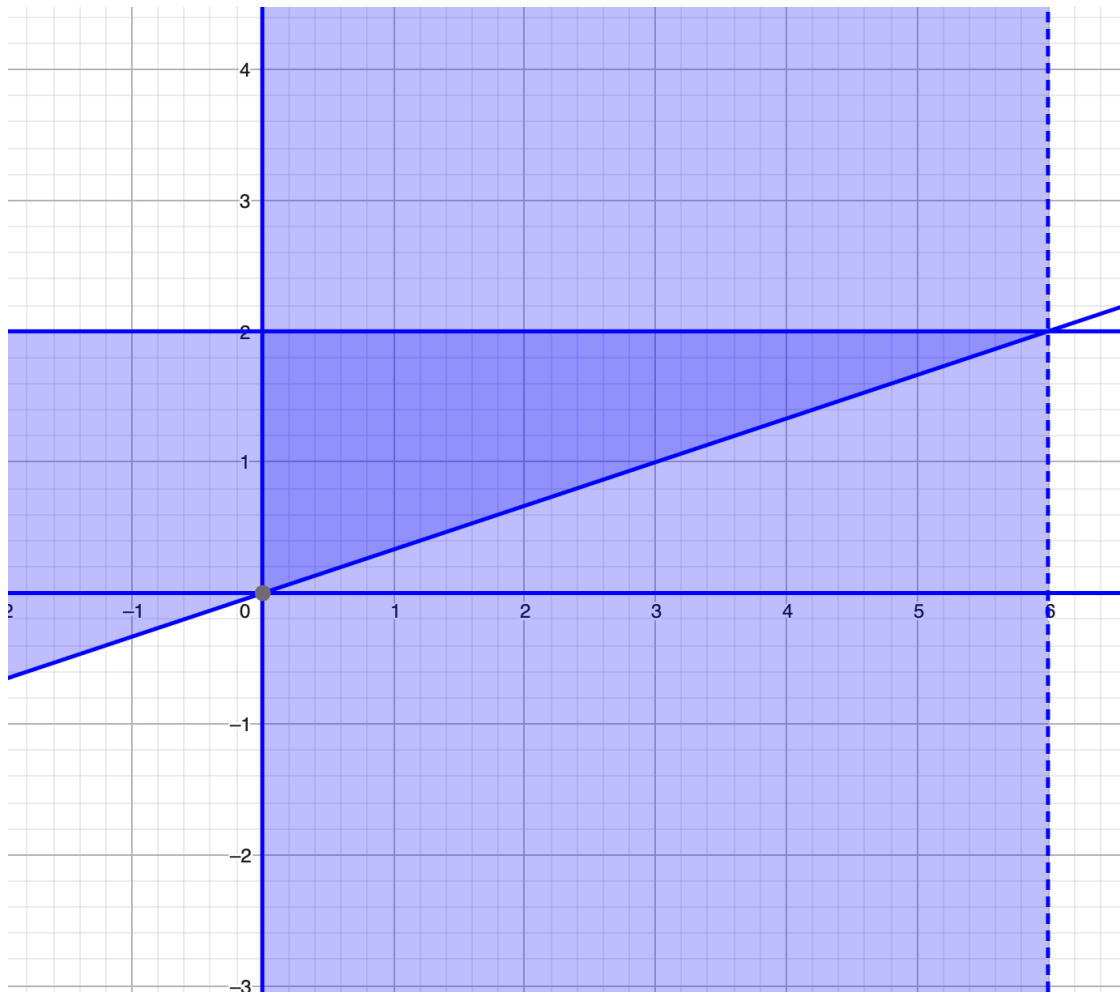
Sketch the region of integration for the iterated integral

$$\int_0^6 \int_{\frac{\pi}{3}}^2 x \sqrt{y^3 + 1} dy dx$$

Then reverse the order of integration.

8.1 Solution

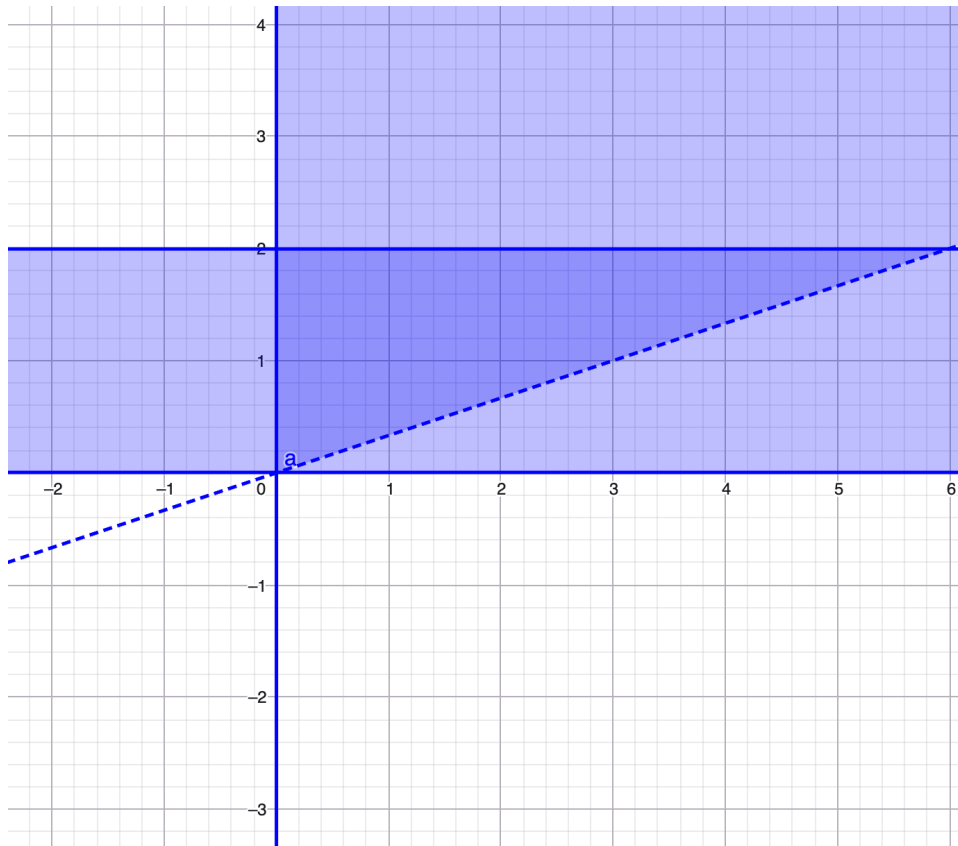
We know that $\frac{x}{3} \leq y \leq 2$ and that $0 \leq x \leq 6$



To reverse the order, first switch the boundary $\frac{x}{3}$ because it depends on x . so...

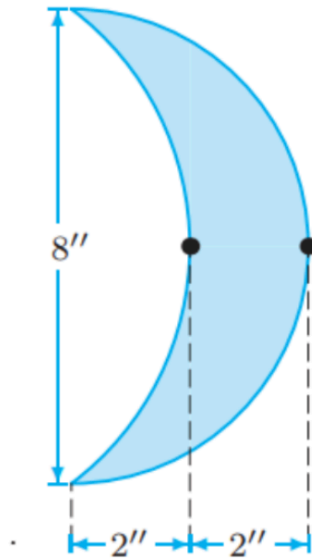
$$y = \frac{x}{3} \Leftrightarrow x = 3y$$

Now we know that $0 \leq y \leq 2$ and that $0 \leq x \leq 3y$



9 Problem 9

Find the area of the crescent-moon shape with circular arcs as edges and the dimensions shown in figure below.



9.1 Solution

We know that

$$A = A_{\text{small}} - A_{\text{segm}}$$

$$A = \frac{1}{2}\pi 4^2 - A_{\text{segm}}$$

So we have to find the segment of the bigger circle

We know that $A_{\text{seg}} = \frac{1}{2}\theta R^2 - \frac{1}{2}ad$. (from)

and since that $R^2 = d^2 + \left(\frac{a}{2}\right)^2 = (R - h)^2 + \left(\frac{a}{2}\right)^2$

$$R^2 = R^2 - 2hR + h^2 + \frac{a^2}{4} \Rightarrow R = \frac{1}{2h} \left(h^2 + \frac{a^2}{4} \right) = \frac{1}{2} \left(h + \frac{a^2}{4h} \right)$$

since $\sin\left(\frac{\theta}{2}\right) = \frac{\frac{a}{2}}{R}$

$$\theta = 2 \sin^{-1} \left(\frac{a}{2R} \right)$$

$$A_{\text{seg}} = \sin^{-1} \left(\frac{ah}{h^2 + \frac{a^2}{4}} \right) \cdot \left(\frac{h}{2} + \frac{a^2}{8}h \right) - \frac{1}{2}a \left(\frac{a^2}{8h} - \frac{h}{2} \right)$$

$$A = \frac{1}{2}\pi 4^2 - 12.68$$